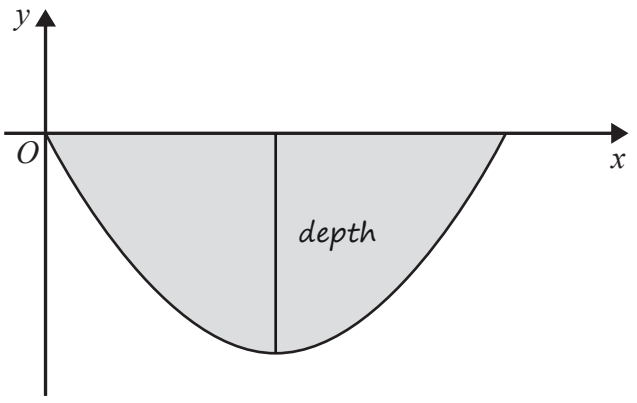
15 Here is a sketch of a vertical cross section through the centre of a bowl.



The cross section is the shaded region between the curve and the x-axis.

The curve has equation $y = \frac{x^2}{10} - 3x$ where x and y are both measured in centimetres.

Find the depth of the bowl.

At the x-axis, y = 0:

$x^2 \div 10 - 3x = 0$, so $x^2 - 30x = 0$.

1 mark
Set y = 0

×	x	-30
x	x^2	$-30x$

$x(x - 30) = 0$

$x = 0$ or $x = 30$.

1 mark
Find both roots

Axis of symmetry: $x = (0 + 30) \div 2 = 15$.

At $x = 15$: $y = 225 \div 10 - 45 = -22.5$.

1 mark
Substitute midpoint
x = 15

Depth = $0 - (-22.5) = 22.5$ cm.

1 mark
22.5 cm

..... cm

(Total for Question 15 is 4 marks)